

# JAVASCRIPT-BASED LISSAJOUS CURVE SIMULATOR

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Cathode-ray tube (CRT) oscilloscopes are generally used in the electronics industry to measure the signal waveforms and also to compare two or more signal waveforms. Most oscilloscopes come with up to six channels of inputs. Each of these signals can be checked individually on an electron-gun CRT oscilloscope. But with the advent of spectroscopes and digital storage oscilloscopes, multi-colour graphical plots can also be viewed and saved for review at a later time.

This JavaScript-based program is used for simulating colourful, animated Lissajous or Bowditch curves. A Lis-

sajous curve is generally observed on a two-channel CRT oscilloscope as it compares 'x' and 'y' inputs. The curve is the depiction of the mathematical system of parametric equations in a graphical form. It is used to calculate the differences in 'x' and 'y' functions simultaneously.

The program displays an animation of the plotting of Lissajous curves at different 'x' and 'y' channel inputs. The program parameters are also displayed along the pattern/curve.

## Software program

The program 'lissajous.htm' written in JavaScript takes the advantage of mathematical library to plot Lissajous



curves directly on the screen. These special curve patterns are generated on a dual-trace (two-channel) oscilloscope when two signals of different time periods, amplitudes and voltages are applied to the two channels. (For further details on this curve, refer [http://en.wikipedia.org/wiki/Lissajous\\_curve](http://en.wikipedia.org/wiki/Lissajous_curve).) Screenshot 1 shows a Lissajous pattern plotted by this software program.

## Function description

The program plots the curves using 'Lissajous()' function that plots two separate curves for each of the inputs—x and y. It then divides the screen area as per the left and top of HTML alignment tags. The left tag describes how far the program should position the plot from the screen's left area. Similarly, the top tag defines the position of the plot on the top of the screen.

The program plots the curve according to the following equations defined in the code:

$$\text{Sc}[i].\text{style.top}=450+320*\text{Math.sin}(d+i*\text{Math.PI}/a);$$

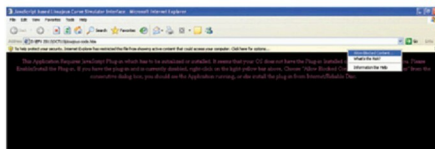
$$\text{Sc}[i].\text{style.left}=650+320*\text{Math.sin}(d+i*\text{Math.PI}/b);$$

Here Sc is a variable-length array that defines the screen area. It is taken as an array because it is not normally known how many variables a function can hold to display the dots (.) on screen.

A 50.8cm (20-inch) LG Flatron E2040T monitor with 1280×1024 resolution is used for testing. The curve is first plotted at 450 pixels from the top



Screenshot 1: Lissajous pattern plotted by the program



Screenshot 2: Windows security warning message